



OPTIMISING IN VITRO TRANSCRIPTION FOR SCALABLE mRNA SYNTHESIS

**Improving Process Control and Reproducibility with
Radleys Reactors**

The development of high-quality biological therapeutics requires efficient and reliable in vitro transcription (IVT). Conventionally, in vitro transcription is performed in batch reactions in single-use setups, which have been shown to exhibit batch-to-batch variation due to fluctuating reaction conditions. Glass reactors are a promising method to address this batch variation issue, and with a leaning protocol in place, the reactors can be used repeatedly. This application note discusses the use of glass jacketed reactors in improving in vitro transcription's efficiency and reliability, particularly for mRNA synthesis.

Keywords or phrases:

mRNA synthesis, in vitro transcription (IVT), Radleys reactors, bioprocess scale-up, process control, RNA therapeutics

INTRODUCTION

In vitro transcription (IVT) is a widely used technique for synthesising ribonucleic acid (RNA) in the laboratory. It involves the use of a deoxyribonucleic acid (DNA) template, RNA polymerase enzyme, and nucleotides to produce RNA molecules of interest.

The reaction conditions in IVT are critical for the efficiency and yield of RNA synthesis. The conventional method of in vitro transcription is performed in batch reactions, which have been shown to exhibit batch-to-batch variation due to fluctuating reaction conditions.

One way to optimise these conditions is through the use of jacketed glass reactors. This application explores the advantages of using jacketed glass reactors in IVT reactions, specifically for mRNA synthesis.

THE ROLE OF mRNA IN THE PHARMACEUTICAL INDUSTRY

The pharmaceutical industry has witnessed a paradigm shift with the advent of mRNA technology. mRNA has emerged as a powerful tool for therapeutic interventions, enabling innovative approaches in personalised medicine and vaccine development. It became a global game-changer during the COVID-19 pandemic—delivering the world's first mRNA-based vaccines at unprecedented speed.

mRNA: A PLATFORM FOR PERSONALISED MEDICINE

Creating treatments that are tailored to each patient's distinct genetic profile holds the promise of personalised medicine, and mRNA is quickly emerging as one of the most flexible platforms for this purpose.

Synthetic mRNA can be modified to encode therapeutic proteins specific to a patient's disease biology through the use of in vitro transcription (IVT). These consist of tumour-specific antigens for cancer immunotherapy, enzymes, antibodies, and immune modulators. Since mRNA therapeutics can be easily modified to encode almost any protein of interest, they are not constrained by target specificity like traditional medications are.

When treating rare or genetically complex diseases, where one-size-fits-all treatments frequently fail, this flexibility is essential. Clinicians can use customised mRNA constructs to target molecular abnormalities by combining genomic profiling with IVT technology, enabling not only precise but also faster development and safer administration.

mRNA VACCINES: PANDEMIC RESPONSE TO BROAD-SPECTRUM PREVENTION

For mRNA technology, the COVID-19 pandemic marked a turning point. The creation, approval, and distribution of mRNA vaccines in less than a year revolutionised vaccine science and international public health policy.

mRNA vaccines provide genetic instructions that cause the body's own cells to produce a disease-specific antigen, in contrast to conventional vaccines that employ inactivated pathogens or protein subunits. Without integrating into the patient's genome or exposing them to live virus, this sets off a focused immune response.

Why mRNA vaccines are revolutionary

Speed: Vaccine candidates can be created in a matter of weeks after a viral sequence is known.

Adaptability: The mRNA sequence can be promptly updated to address new variants.

Safety: The mRNA doesn't change the host DNA and is only temporary.

mRNA-based vaccines against influenza, RSV, HIV, and emerging zoonotic diseases have become widely available as a result of the success of the Pfizer-BioNTech and Moderna COVID-19 vaccines, which are both based on this technology.

mRNA platforms are positioned to play a key role in vaccine development, not only for infectious diseases but also possibly for cancer and chronic conditions, as the focus of the world moves from crisis response to proactive prevention.

CHALLENGES IN mRNA SYNTHESIS

mRNA therapies still have a long way to go before they are widely used in clinical settings, despite their revolutionary potential. Making sure that mRNA reaches the appropriate cells and is effectively translated into useful protein is still a major challenge in targeted delivery.

- ▶ **Batch-to-batch variability** due to inconsistent temperature or mixing

- ▶ **Enzyme deactivation** from local overheating or poor pH stability
- ▶ **Incomplete transcription** leading to short or truncated RNA species
- ▶ **Residual DNA or impurities** interfering with downstream purification
- ▶ **Low reproducibility** when scaling up using single-use or open systems

IN VITRO TRANSCRIPTION

A laboratory technique called in vitro transcription (IVT) is used to create RNA molecules from a DNA template outside of living cells. By using RNA polymerase to create RNA strands that carry genetic instructions, it mimics the natural transcription process.

A DNA template with a promoter sequence that the polymerase recognises (typically T7, SP6, or T3) is used to start the process. This PCR or cloning-prepared template is incubated with:

- ▶ RNA polymerase
- ▶ Ribonucleotides (ATP, GTP, CTP, and UTP)
- ▶ Essential cofactors and buffer

Under regulated circumstances, RNA polymerase attaches itself to the promoter and adds ribonucleotides in a specific order to create a complementary RNA strand. Until a predetermined endpoint or termination signal is reached, the reaction keeps going.

IVT is a foundational step in the production of mRNA therapeutics, vaccines, guide RNAs (CRISPR), and RNA-based assays—offering precision, flexibility, and scalability for modern molecular biology.

REACTOR-BASED SOLUTIONS FOR IVT

Because of their scalability, temperature control, and chemical resistance, glass jacketed reactors

are being used more and more for mRNA in vitro transcription (IVT). These reactors, which are made of inert, corrosion-resistant glass, are perfect for the production of biopharmaceuticals, where process consistency and sterility are crucial. To guarantee even mixing, which is crucial for preserving enzyme activity and reaction homogeneity during RNA synthesis, they are outfitted with a variety of agitator types.

ADVANTAGES OF REACTOR SYSTEMS:

- ▶ Stable temperature control (typically ± 0.5 °C)
- ▶ Uniform mixing to avoid local substrate or enzyme gradients
- ▶ Real-time monitoring of pH, temperature, and process kinetics
- ▶ Efficient RNA recovery and sample handling
- ▶ Scalable volumes, from millilitres to tens of litres

RADLEYS SYSTEMS FOR CONTROLLED IVT

mRNA therapies still have a long way to go before they are widely used in clinical settings, despite their revolutionary potential. Making sure that mRNA reaches the appropriate cells and is effectively translated into useful protein is still a major challenge in targeted delivery.

Mya 4 Reaction Station

- ▶ Ideal for developing new methods
- ▶ 4 Reaction Station (2–400 mL)
- ▶ Data logging, heating/cooling, and stirring automation
- ▶ Allows for multiple parallel reactions to optimise the process.

Reactor-Ready & Reactor-Ready Pilot

- ▶ Glass jacketed vessels (100 mL to 35 L)
- ▶ Using circulators to precisely regulate the temperature
- ▶ Simple vessel swapping and tool-free assembly
- ▶ Facilitates sampling, filtration, and integration with inline probes.

KEY CONSIDERATIONS FOR REACTOR-BASED IVT

Parameters	Impact on IVT
Temperature	Affects RNA polymerase activity and product integrity
pH control	Stabilises enzyme function, reduces degradation
Mg ²⁺ and buffer systems	Support efficient NTP incorporation
Mixing	Prevents hot spots or localised gradients
Volume scalability	Allows same conditions to be maintained from lab to pilot scale
Cleanability & validation	Essential for GMP-readiness and multi-batch operation

CASE STUDY: The Scale-Up Success of Aldevron

Radleys' Reactor-Ready Pilot system was used by Aldevron, a top producer of therapeutic proteins, mRNA, and plasmid DNA, to support their expanding RNA production capacity.

They needed a platform that could:

- ▶ Allow for different batch sizes.
- ▶ During multi-hour IVT runs, maintain strict thermal control.
- ▶ Encourage the integration of quality systems and purification workflows.

Aldevron was able to transition from lab-scale to pilot production without sacrificing RNA integrity, yield, or reproducibility thanks to its robust stirring, tool-free operation, and scaling capability from 100 ml to 35 L.

CONCLUSION

Consistent, scalable, and strictly regulated in vitro transcription procedures are more important than ever as the need for high-quality mRNA in biopharma and vaccine development keeps growing. Precision and reproducibility are frequently lacking in traditional batch methods, particularly when scaling up.

Researchers and manufacturers can overcome important obstacles like temperature drift, mixing inefficiencies, and batch-to-batch variability by incorporating Radleys reactors into the IVT workflow. Radleys systems provide a stable, adaptable platform that connects laboratory research and development with pilot-scale manufacturing thanks to their accurate thermal control, real-time monitoring, and scalable volumes.

In the end, this method facilitates the dependable and GMP-compliant production of next-generation RNA therapies in addition to streamlining mRNA synthesis.

REFERENCES

<https://www.radleys.com/resources/in-vitro-transcription-ivt-in-jacketed-glass-reactors/>

<https://www.radleys.com/case-studies/aldevrons-use-of-radleys-reactor-ready-pilot-for-mrna-production/>



MYA 4



REACTOR READY

